Grokking Under a Fixed Budget

One long run or several restarts?

Probability and Statistics
Final explanation and measured results

Name: [Your name]
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Grokking explained with an example

Memorization: answering familiar practice examples correctly.

Generalization: answering questions held back from practice.

Grokking: new-question accuracy improves after a delay.

Our rule: (30 + 4) mod 31 = 3. Modulo keeps the remainder.

The computer sees number pairs and learns from examples.

The question and experiment

With the same total steps, which policy succeeds more often?

961 pairs: 576 for practice and 385 for evaluation.

One small model. Three fixed dataset splits.

A seed fixes starting weights. Weight decay gently shrinks them.

90 calibration observations estimate learning-time probabilities.

540 fresh attempts test 1, 2 or 4 starts at 8k and 12k total steps.

How the code works

Python generates the pairs and trains the model.

One update = one learning step using all practice pairs.

Every 25 updates: measure accuracy and save CSV records.

Checkpoints let interrupted training resume.

R computes statistics and creates the graphs.

The extension completed 1,330 new runs and 7.5 million logical updates.

The probability and statistics

T = third good check. U = min(T, C). C = run limit.

Good check: at least 99% practice accuracy and 95% new-question accuracy.

$$P(\text{at least one success}) = 1 - [1 - F(B/k)]^k$$

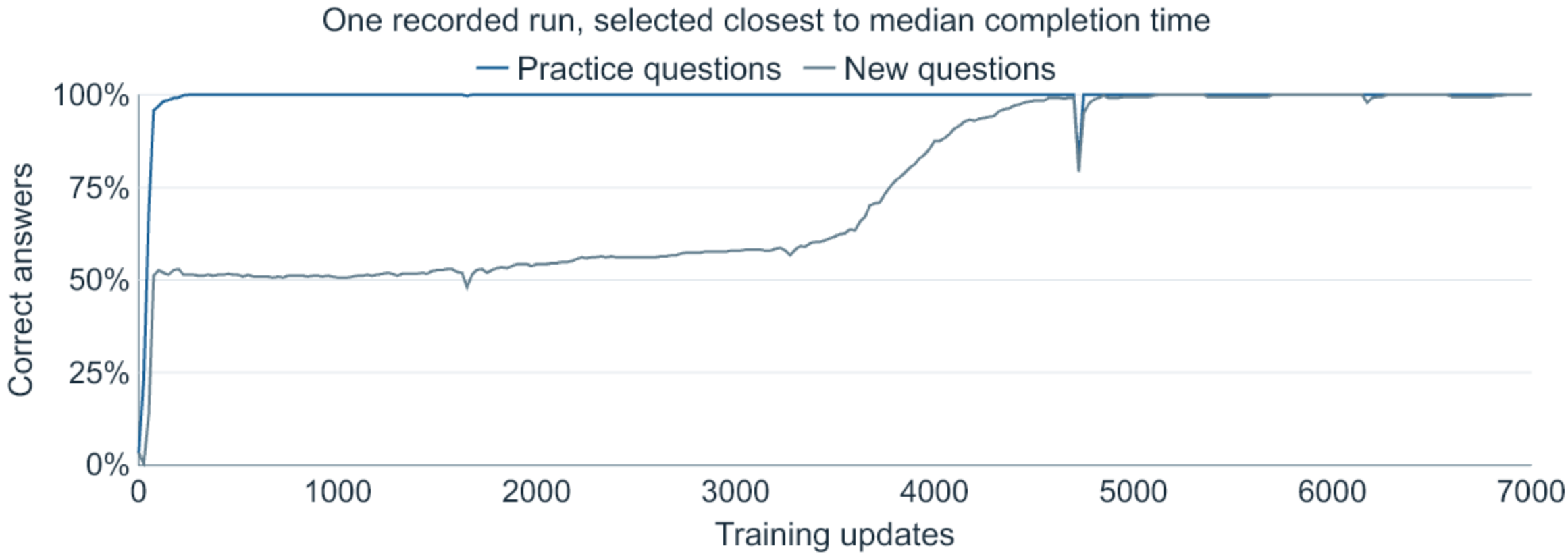
B = total steps. k = starts. F(t) = chance one run succeeds by t.

Bernoulli / binomial: yes-or-no attempts and success counts.

Exact 95% intervals describe uncertainty.

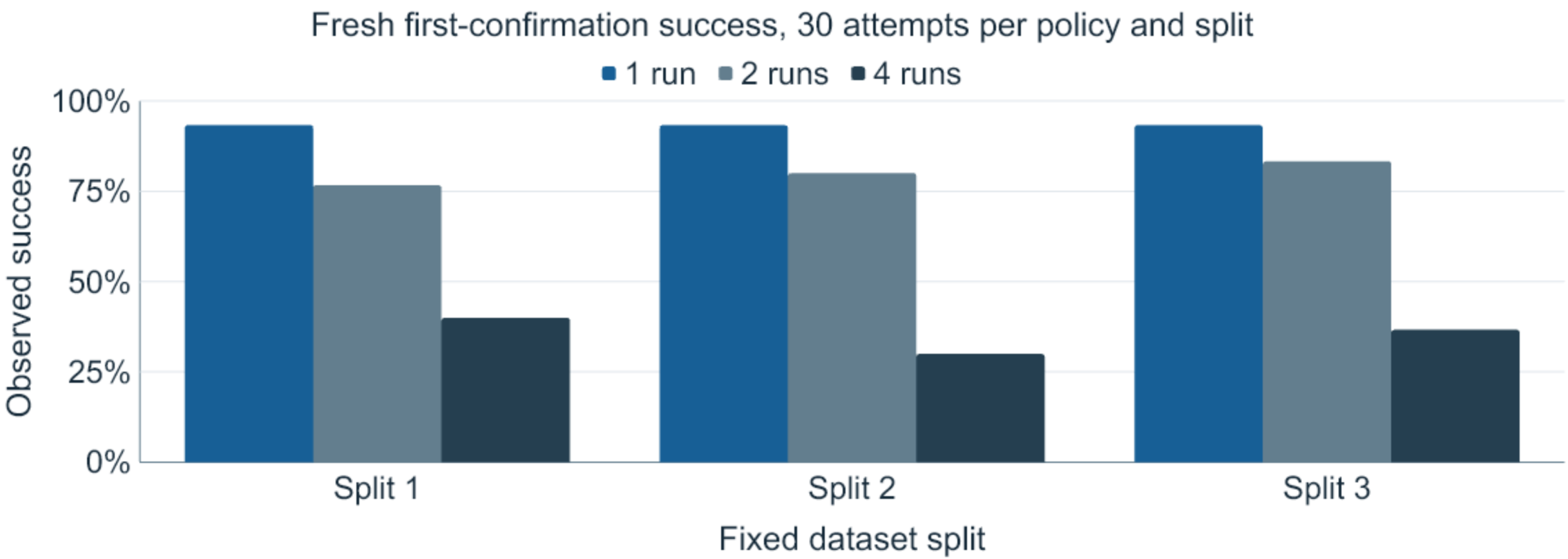
Twelve Fisher tests with Holm correction compare the policies.

Practice improves before new-question accuracy



Original-split mean: 5,347 updates. Median: 4,375. SD: 4,610 (n = 30).

Results at 8,000 total updates



1 run: 28/30 on every split. 4 runs: 12/30, 9/30, 11/30.

All three 4-versus-1 differences survive Holm correction. No 2-versus-1 contrast does.

Complete results at both budgets

Split / total steps	One run	Two runs	Four runs
S1 / 8,000	28/30	23/30	12/30
S1 / 12,000	30/30	30/30	24/30
S2 / 8,000	28/30	24/30	9/30
S2 / 12,000	30/30	30/30	20/30
S3 / 8,000	28/30	25/30	11/30
S3 / 12,000	30/30	29/30	23/30

At 12k, only the split-2 four-run comparison survives correction.

Example uncertainty: 28/30 = 93.3%, exact 95% interval 77.9% to 99.2%.

First success and longer stability

First confirmation needs 3 good checks.

The 500-update window needs 21 good checks within the budget.

Split 1 at 8k, four starts: 12/30 first, but 4/30 longer-window successes.

Neither endpoint guarantees that accuracy stays high forever.

Limits: one task, one model and three splits. Small differences remain uncertain.

Future work: other tasks, more splits and longer accuracy windows.

Conclusion

At 8,000 total updates, one long run was more reliable than four short runs on all three splits.

Two-run comparisons remain inconclusive. Nonsignificance does not prove equality.

Established ideas credited. Our contribution is the recorded empirical comparison.

Power et al., Grokking, arXiv:2201.02177 (2022).
Streeter, Golovin and Smith, Restart Schedules for Ensembles of Problem Instances (2007).
Full references and all uncertainty intervals: Assessment 9 report.